

# Algorithmic Trading Strategy Development on Multi-Feature Time Series



**EBULLIENT  
SECURITIES**



**INTER IIT  
TECH MEET 14.0**

Team 33

|              |    |           |           |           |       |
|--------------|----|-----------|-----------|-----------|-------|
| REUTERS      |    |           |           |           |       |
| 555750       | 9  | 15,040 €  | 15,041 €  | 15,041 €  | 1,35% |
| 833212       | 10 | 12,648 €  | 12,648 €  | 12,648 €  | 1,35% |
| tsche 843002 | 3  | 176,020 € | 176,004 € | 176,042 € | 1,35% |
| 510000       | 23 | 88,864 €  | 88,870 €  | 88,871 €  | 0,01% |
|              |    |           | 88,869 €  | 88,861 €  | 0,01% |



# Problem Statement & Core Objectives

- The challenge required the development of a single, robust **intraday trading strategy** for two distinct, high-frequency datasets: **EBX and EBY**. A central requirement was that the same core logic must demonstrate consistent performance across both markets, ensuring its adaptability and preventing overfitting.
- The strategy had to operate under strict constraints, including a **mandatory end-of-day flat position** and the absence of any forward bias in its signals.
- Performance was evaluated primarily by the **Calmar Ratio**, making the **maximization of risk-adjusted returns** the ultimate objective.

### Noisy Price Data

Both datasets were characterized by **short term price fluctuation**. This noise was particularly pronounced in EBY, making the extraction of true directional signals a primary obstacle.

# Exploring Market Dynamics

### Whipsaws

The data showed clear shifts between **trending** and **choppy, consolidating** phases. The ability to correctly identify the current market regime was deemed critical to avoid losses from "**whipsaws**" in sideways markets.

### Buy and Hold Return

A simple buy-and-hold strategy yielded a **net-zero return** over the entire period. This confirmed that profitability could only be achieved through active intraday trading, not by betting on a long-term direction.

### Outliers

While average intraday price swings were **1-1.5%**, dataset featured outlier days with massive volatility. For instance, **Day87 (EBX)** and **Day104 (EBY)** saw extraordinary **8-10%** swings, highlighting the necessity for robust risk management.





Experimented  
Approaches

Trend Classification With MA Ribbon And HA Candles

Trend Filtering With KAMA And CHOP

Variational Autoencoder And XGBoost

Reinforcement Learning



# Trend Classification

## Mechanism 1: Heikin Ashi Smoothing

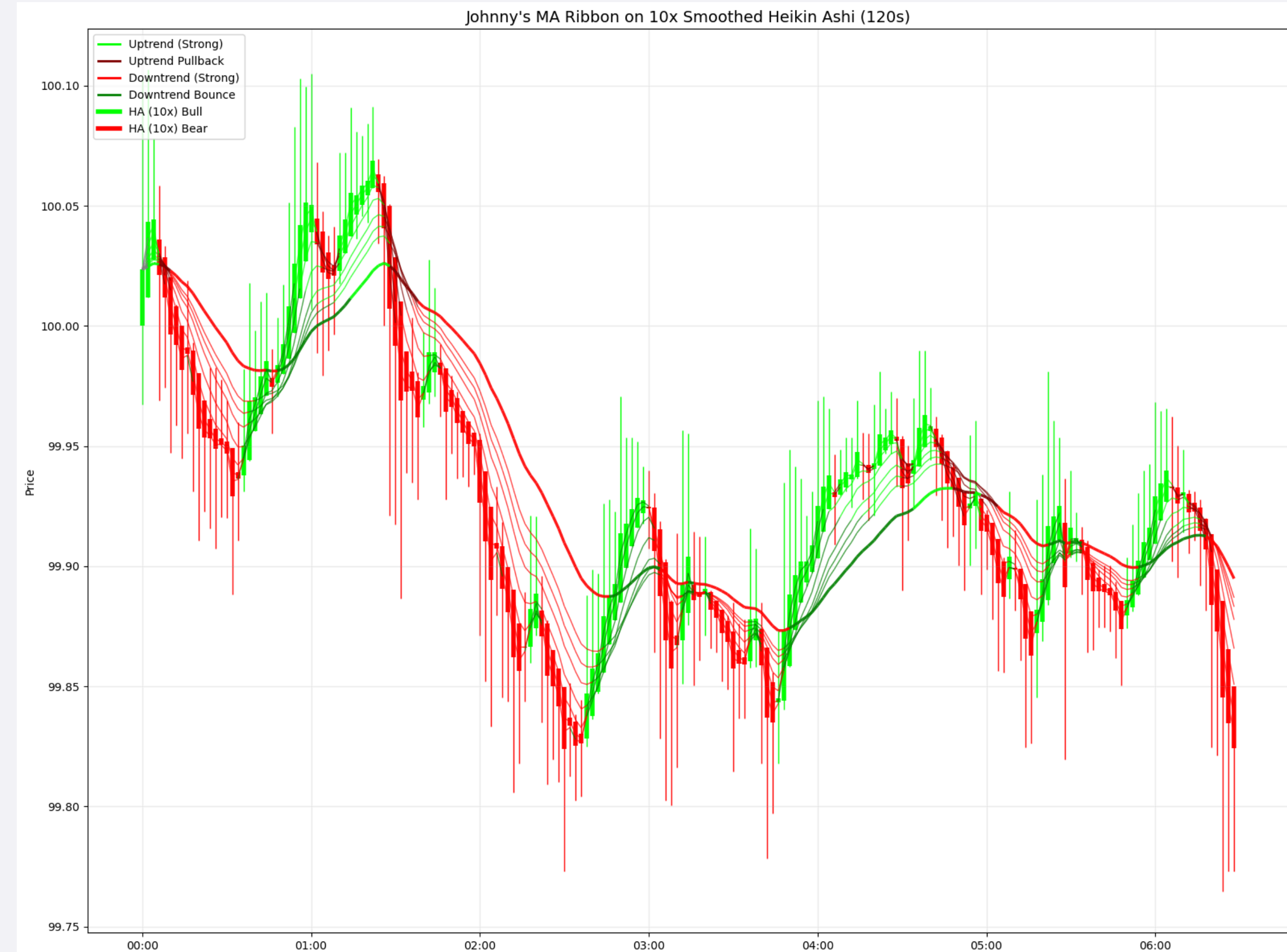
- Transforms standard price data to **reduce volatility** and **highlight trend persistence**.
- HA-Close: Average of the current bar.
- HA-Open: Recursive midpoint of the previous HA body.

$$HA_{Close} = \frac{O + H + L + C}{4} \quad HA_{Open} = \frac{HA_{Open(prev)} + HA_{Close(prev)}}{2}$$

## Mechanism 2: Moving Averages Ribbon

Uses relative positioning of multiple MAs w.r.t a reference MA (MAref).

- Strong Uptrend:** MAs fanned out in ascending order.
- Uptrend Pullback:** MAs > MAref but disordered.
- Strong Downtrend:** MAs fanned out in descending order.
- Downtrend Bounce:** MAs < MAref but disordered.



**Whipsaw Failure:** The rigid classification failed in sideways markets, interpreting noise as trend reversals.

# Trend Filtering

## Mechanism 1: Choppiness Index (The Gatekeeper)

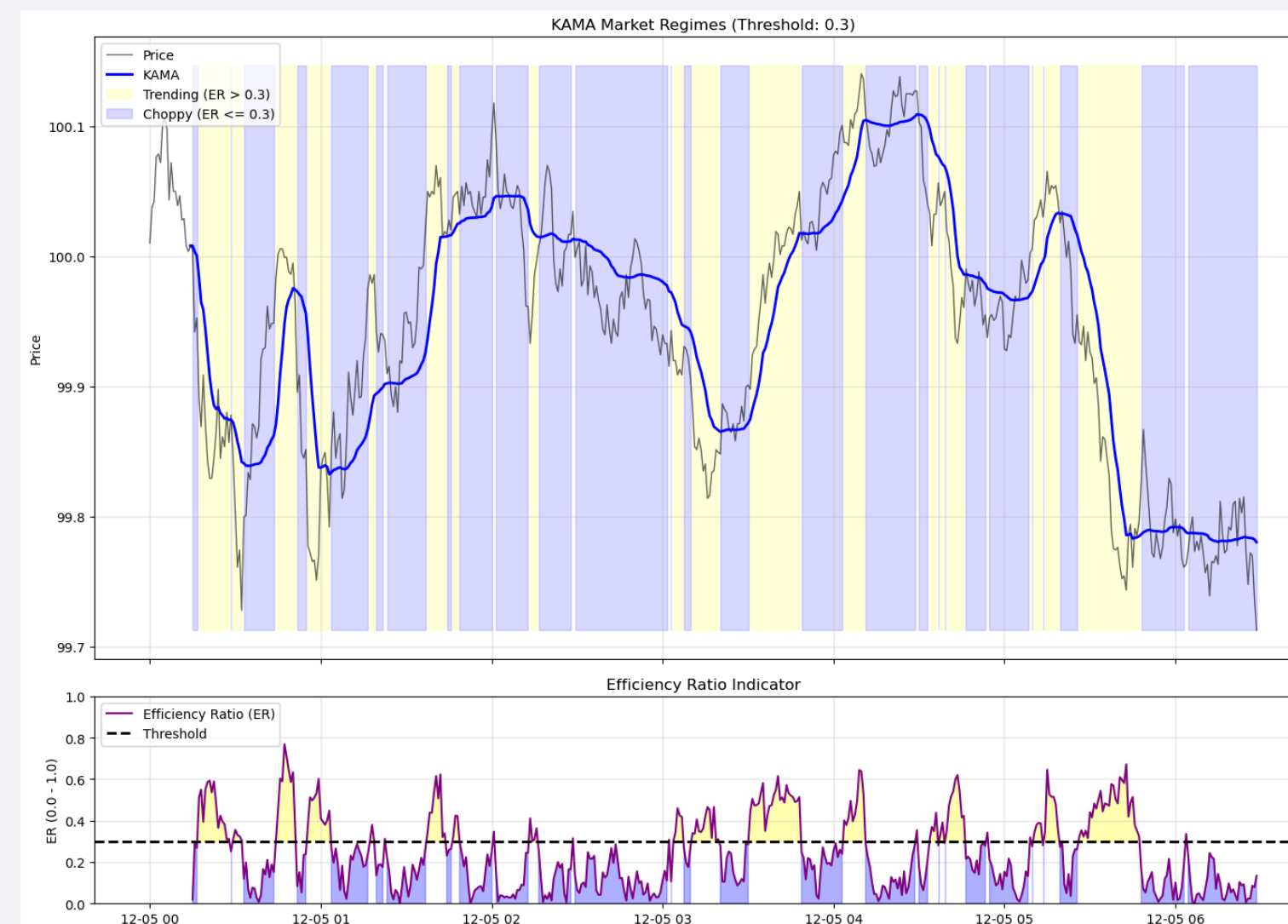
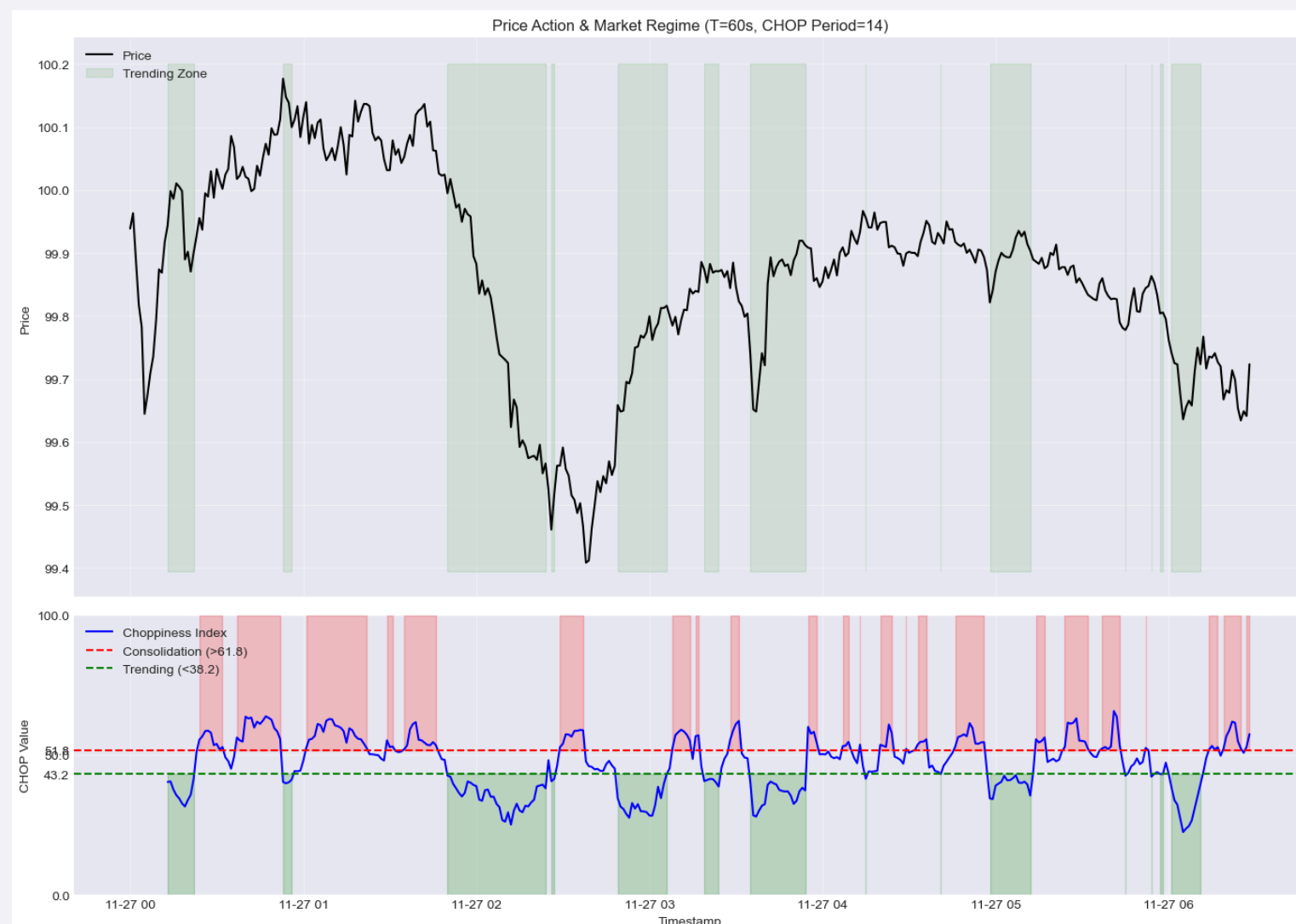
- Measures trendiness using Fractal Dimension logic.
- Trades only permitted when CHOP < 43.2.

$$CHOP = 100 \cdot \frac{\log_{10}(\sum ATR) - \log_{10}(MaxH - MinL)}{\log_{10}(Period)}$$

## Mechanism 2: KAMA (The Adaptive Signal)

- Efficiency Ratio (ER): Determines signal-to-noise quality (1 = Trend, 0 = Noise)
- Recursive Calculation: KAMA adapts its speed (SC) based on ER.

$$ER = \frac{|Price_{change}|}{\sum |Price_{fluctuations}|} \quad KAMA_t = KAMA_{t-1} + SC_t \cdot (Price_t - KAMA_{t-1})$$



## Limitations

- Noisy and Lagging:** The features still had considerable noise. Trade-off between noise-accuracy and lag has to be balanced.

## Importance

- Feature Validation:** Proved ER and CHOP are robust feature engineering components for Machine Learning despite the noise.



# VAE-XGBoost

## Variational Autoencoder (VAE)

1. VAE is a generative neural network model that learns to compress high dimensional data into a low-dimensional latent space.
2. Advantageous because of its strong noise reduction feature by sampling and regularization.

### Application in Strategy:

- In addition to features given in dataset, features derived from Heiken-ashi, Moving Averages, Aroon up/down, KAMA and CHOP, were also added.
- VAE compresses the features into a 64 dimensional latent vector.
- After XGBoost predicts Bull/Bear/Sideways probabilities, ER filters trades in trending regions for position sizing.
- High-probability signals ( $\geq 90\%$ ) - maximum allocation (100 units)
- Lower-confidence signals - scaled down (40 and 30 units)

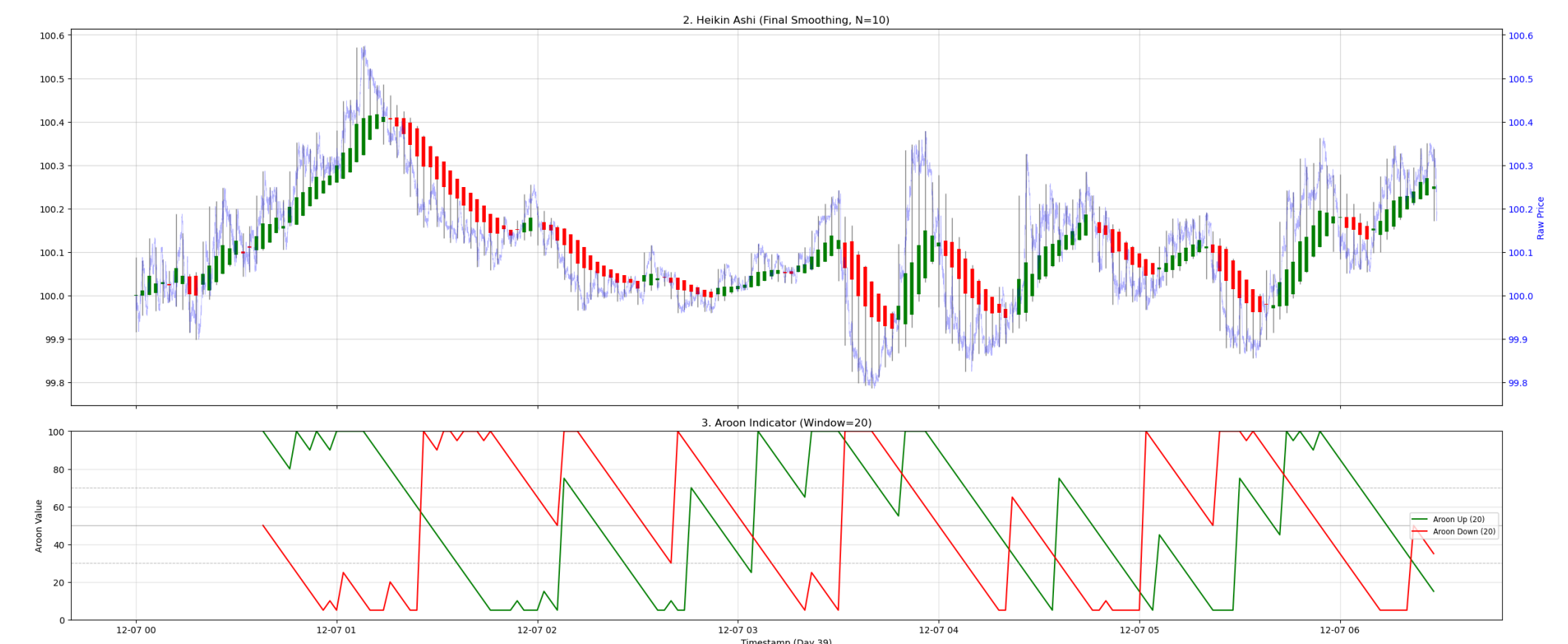
### Limitations of this Strategy:

1. **In the case of high-frequency candles (1-2 minutes):** Noise in choppy regions led to false trades despite ER filtering, due to low prediction confidence.
2. **In the case of low-frequency candles (5-10 minutes):** Insufficient price swings between candle closes limited profitability. Inconsistency between per-second price trends and candle trends reduced accuracy, causing false trades.

## Model predictions and their corresponding confidence

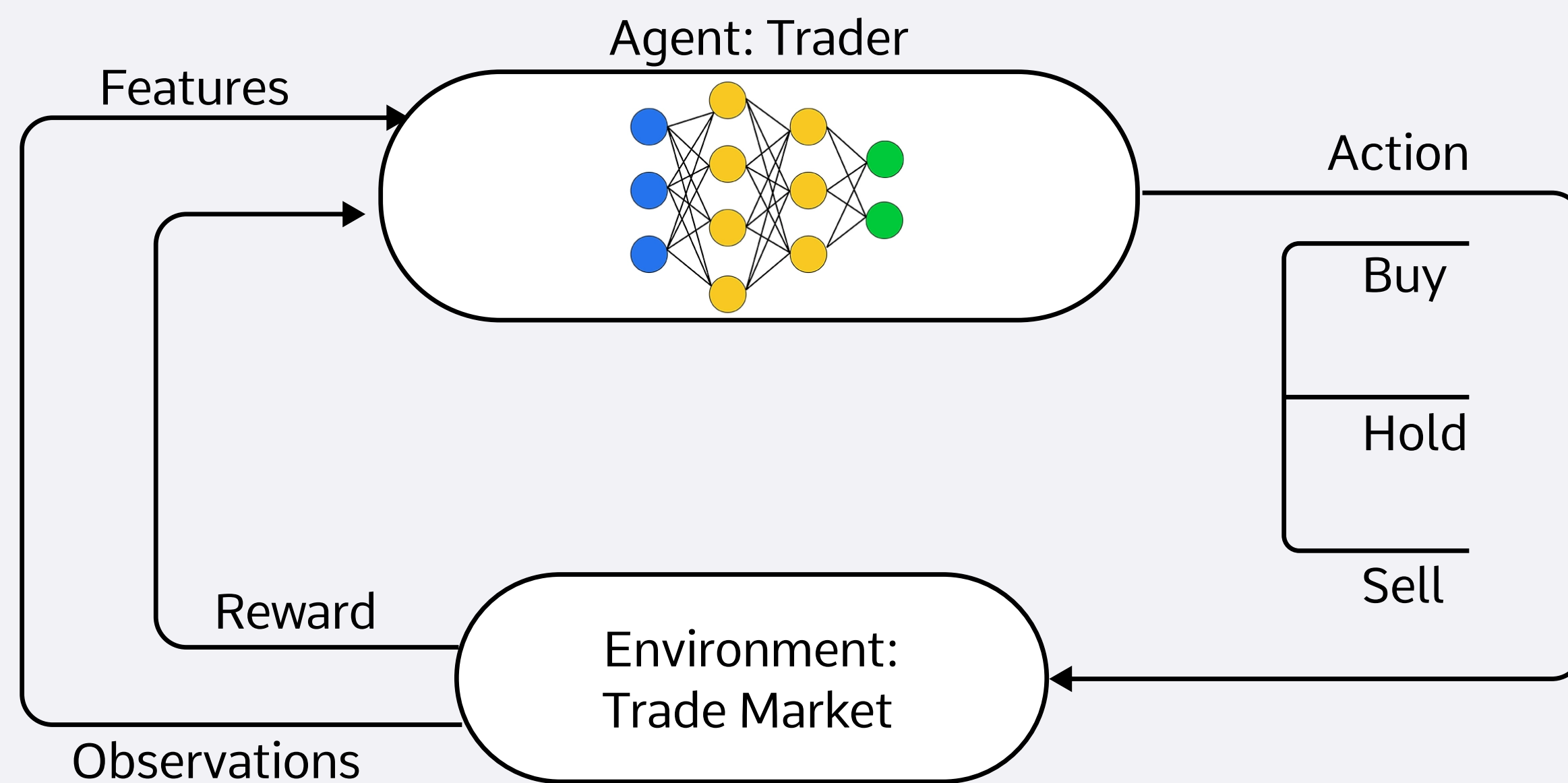


## Aroon Up and Down Indicator



Our Approach:

# Reinforcement Learning (PPO) for Intraday Trading



## Why RL?

- 01 Sequential Decision-Making (2-min steps)
- 02 Directly optimizes trading objective
- 03 Works with complex, high-dimensional signals
- 04 Risk management baked into learning
- 05 Reduces overtrading via reward shaping
- 06 Generalizes across market regimes

# Actor-Critic

An Actor-Critic (AC) model learns a trading strategy through a continuous cycle of **trial-and-error** in a market environment.

The Objective is to find Optimal policy maximizing the reward and the Value function supporting its accuracy

## The Actor Network (The Policy $\pi$ )

- The Actor is a neural network that represents the agent's policy.
- Its job is to decide the action (buy, sell, hold) based on current market state.
- Based on Critic's feedback it learns by increasing probability of actions with positive reward and decreasing probability of actions with negative reward.

## The Critic Network (The Value Function)

- The Critic is a separate neural network that estimates the value function  $V(s_t)$ .
- Its job is to evaluate the quality of the current market ( $s_t$ ) by estimating the expected reward.
- Critic calculates advantage which measures how much better the received reward was than the Critic's estimate.
- Critic updates its parameters to predict closer to the actual reward.

## The Entropy Term

- The Actor might quickly assign a near-zero probability to all but the single best-performing action observed so far.
- The Entropy spreads the probability mass across its available actions, thereby fulfilling the role of exploration regularization

$$H(\pi_\theta(s)) = - \sum_{a \in A} \pi_\theta(a|s) \cdot \log(\pi_\theta(a|s))$$

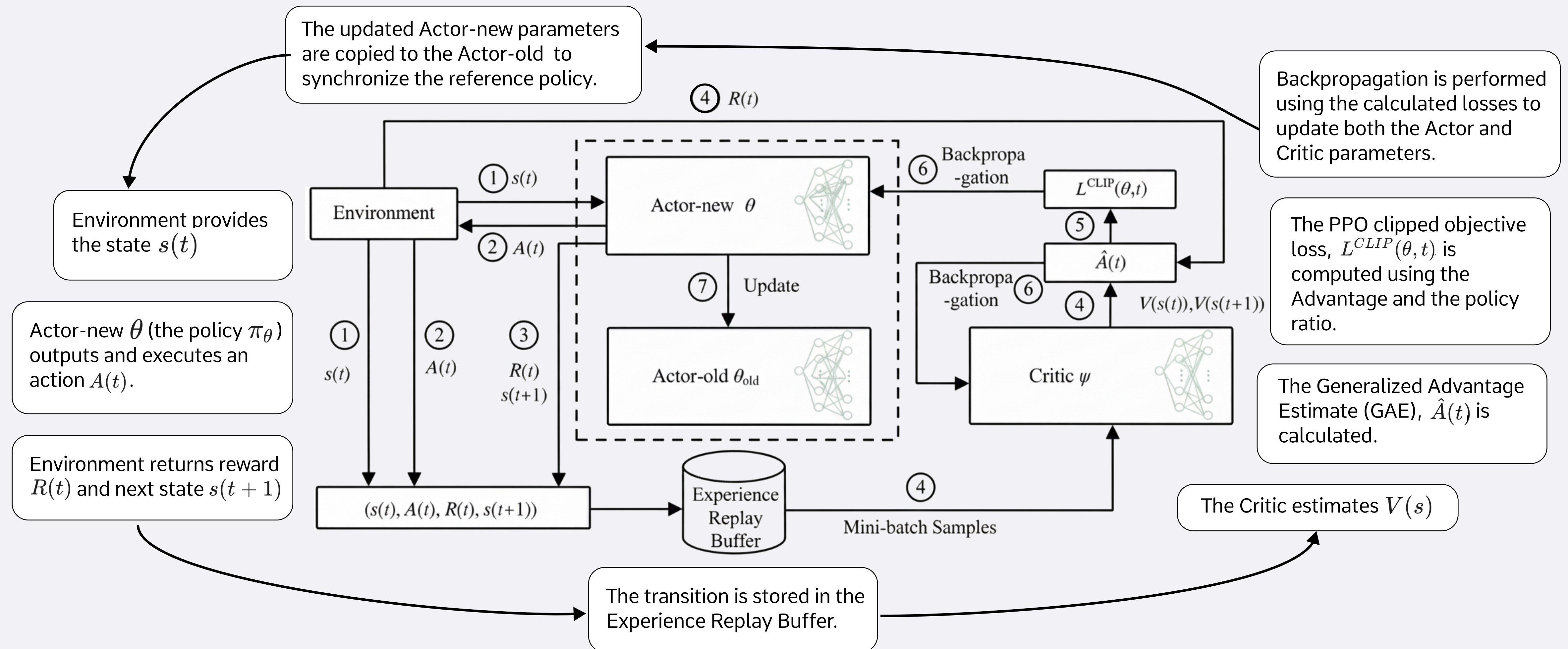
## Objective:

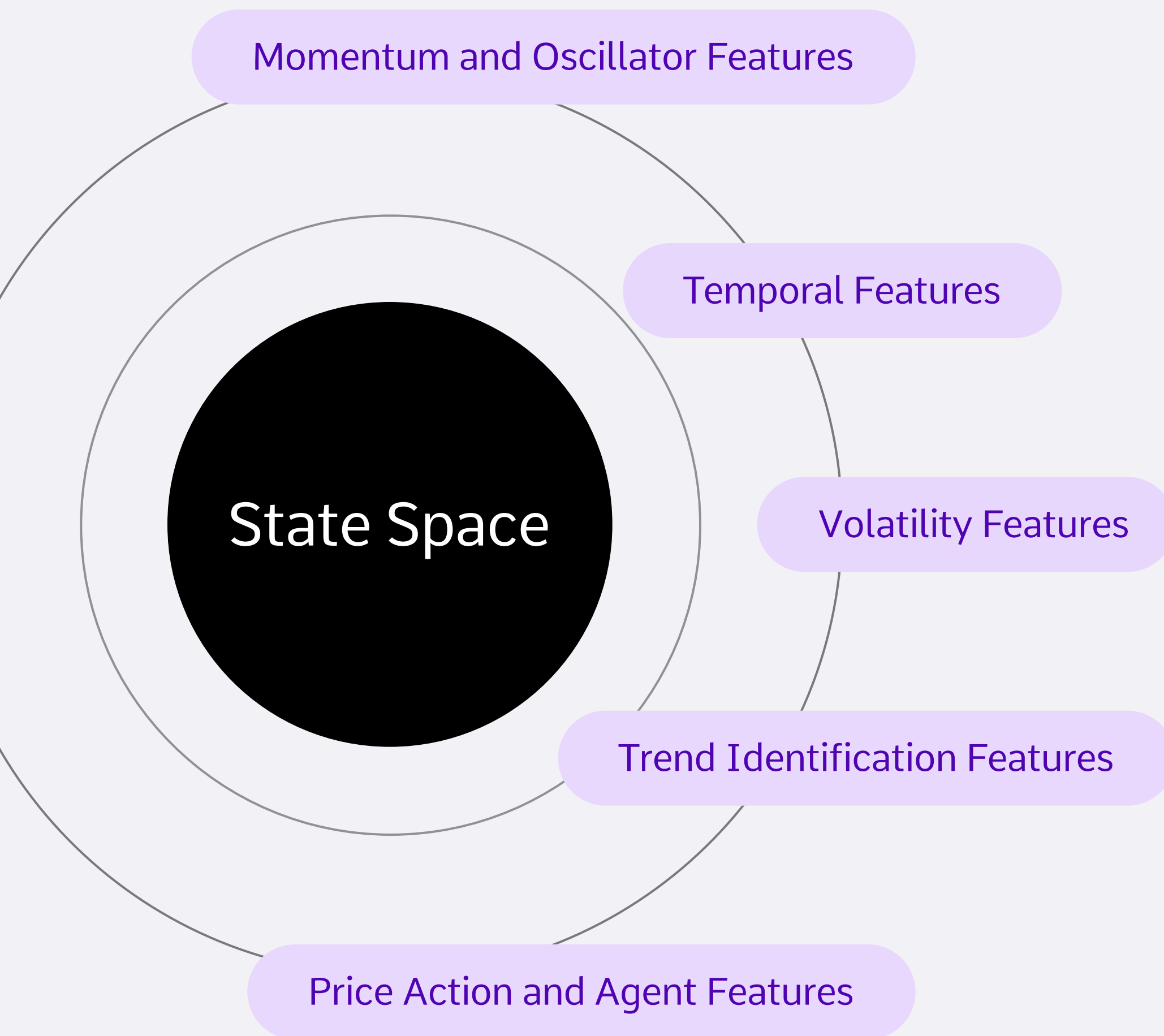
$$\text{Maximize}_\theta \quad L_{\text{Policy}}(\theta) = \hat{\mathbb{E}}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) + c \cdot H(\pi_\theta) \right]$$

$$\text{Minimize}_\psi \quad L_{\text{Value}}(\psi) = \hat{\mathbb{E}}_t \left[ \left( V_\psi(s_t) - V_t^{\text{target}} \right)^2 \right]$$



# PPO: Proximal Policy Optimization





**Note:** Trading decisions and all associated features were computed based on 2-minute candlestick data

### Price Action and Agent Features

We compute **logarithmic returns** of the previous candle's OHLC prices. We also included the agent's internal state, namely the position  $\in \{1, 0, -1\}$  and mark-to-market P&L.

$$r^{(t)} = \ln \left( \frac{P^{(t-1)}}{P^{(t-2)}} \right), \text{ where } P \text{ is the price}$$



### Momentum and Oscillator Features

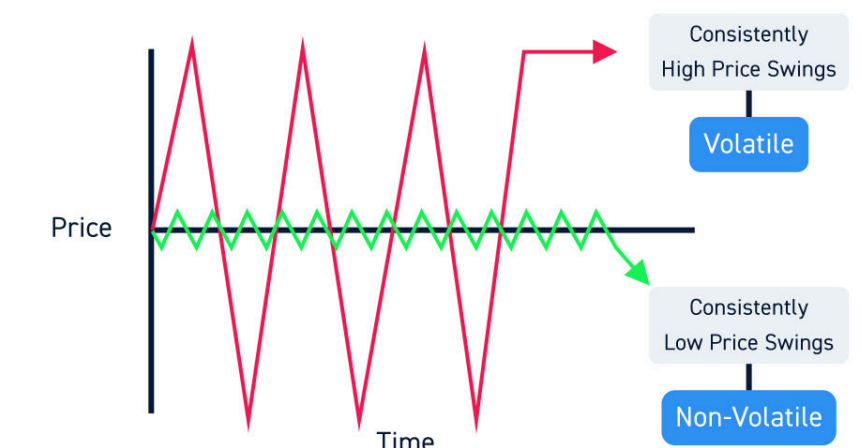
RSI, CMO and CCI calculated over the lookback periods  $\{3, 5, 10, 20\}$  candles :

- RSI indicates overbought and oversold conditions
- CCI identifies cyclical turns and extreme prices
- CMO distinguishes between trending and mean-reverting regimes.

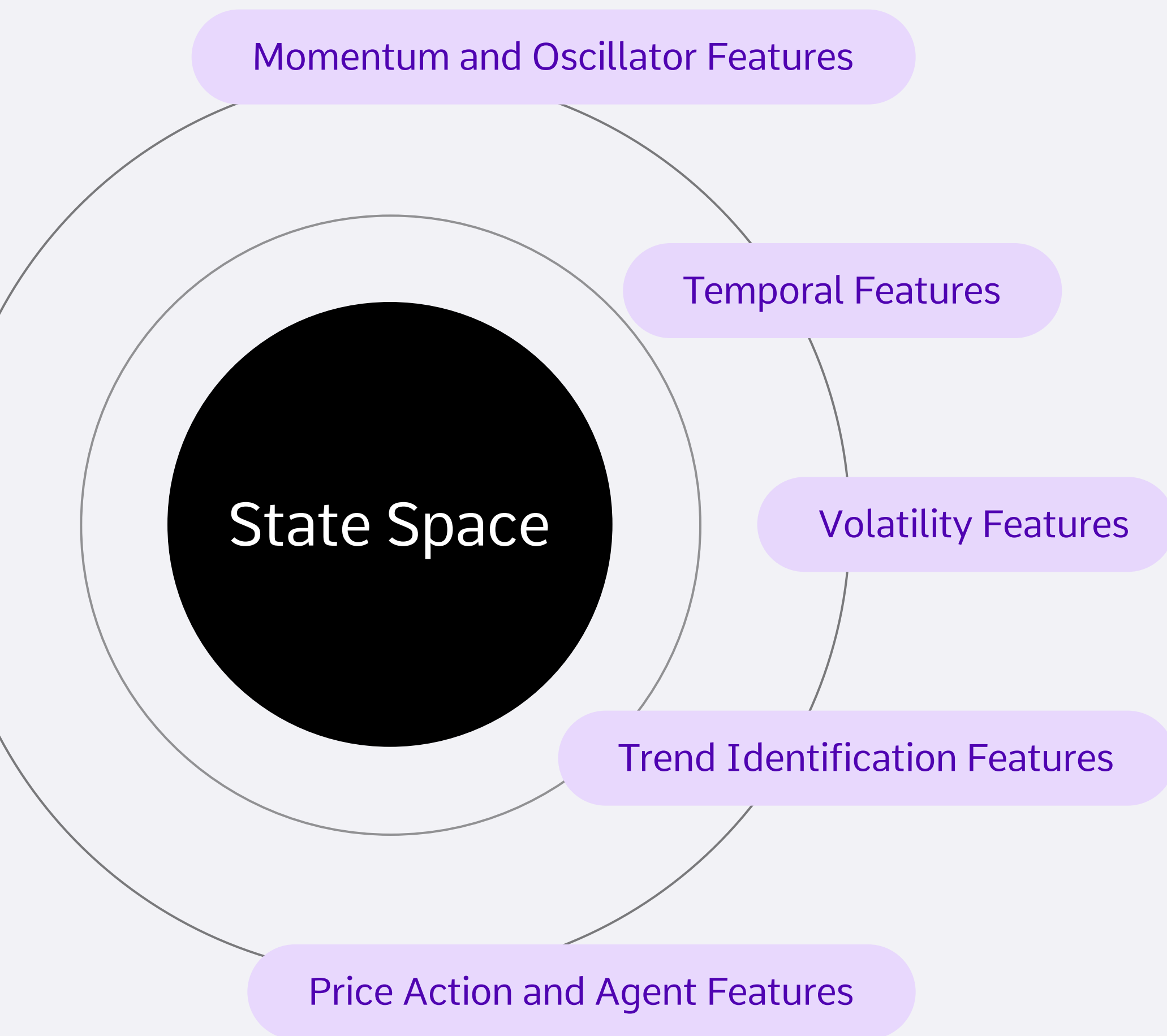


### Volatility Features

Features like **ATR** and **Rolling standard deviation** enable the agent to distinguish between low-volatility periods and high-volatility trending periods.







**Note:** Trading decisions and all associated features were computed based on 2-minute candlestick data

### Trend Identification Features

Heiken-Ashi features like its **trend**, **body-width** and **wick size**.

Trends from moving averages of Heiken-Ashi close price of different look-back periods  $p \in \{2, 3, 5, 8, 12, 15, 18\}$  were also included as encoded features

**Aroon crossovers** signal trend changes – extreme values of Aroon up/down indicate strong directional conviction.

**KAMA, CHOP and ER** for classifying market into trending and sideways



### Temporal Features

Intraday patterns exhibit significant time-of-day effects. We **encode time using cyclical transformations**

$$H_{\sin} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad H_{\cos} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad M_{\sin} = \sin\left(\frac{2\pi \cdot \text{minute}}{60}\right), \quad M_{\cos} = \cos\left(\frac{2\pi \cdot \text{minute}}{60}\right)$$



# Reward Function

The reward function guides the RL agent's behaviour by assigning:

- **Gains** for profitable exits
- **Penalties** for losses, missed opportunities and excessive trading.

## Final Reward Composition

$$R(t) = r_{pnl} + r_{trail} + r_{stop} + r_{opportunity} + r_{entry} + r_{eod}$$

### Profit & Loss (PnL) Reward:

When the agent closes a position, it receives a reward proportional to the realized return. Losses were penalized more heavily than profits .

$$r_{pnl} = \begin{cases} \text{pnl}_{bps} & \text{if } \text{pnl}_{bps} > 0 \\ \lambda_{loss} \cdot \text{pnl}_{bps} & \text{if } \text{pnl}_{bps} \leq 0 \end{cases}$$

**Stop-Loss hit Penalty:** For risk management we add this penalty to discourage agent from taking trades with adverse price movements.

- **Trailing Stop loss**

$$r_{trail} = -10$$

- **Static Stop loss**

$$r_{stop} = -100$$

**Missed Opportunity Penalty :** To encourage the agent to participate in significant market movements, we penalize remaining flat during volatile periods and missing profitable trades.

$$\Delta_w = \frac{|p(t) - p(t - w)|}{p(t - w)}, \quad w = 5$$

$$r_{opportunity} = \begin{cases} \beta_{miss} & \text{if } \Delta_w > \theta_{opp} \\ \beta_{wait} & \text{if } \Delta_w \leq \theta_{opp} \end{cases}$$

**Trade Entry Penalty:** To discourage excessive trading frequency we apply a fixed penalty upon entering any new position.

$$r_{entry} = -5$$

**End-of-Day Penalty :** To encourage the agent to close trades before the trading session ends.

$$r_{eod} = -10$$

# Results and Key Metrics for EBX

**81.61%** Annualized Returns

**1.15%** Max Drawdown

**70.96** Calmar Ratio

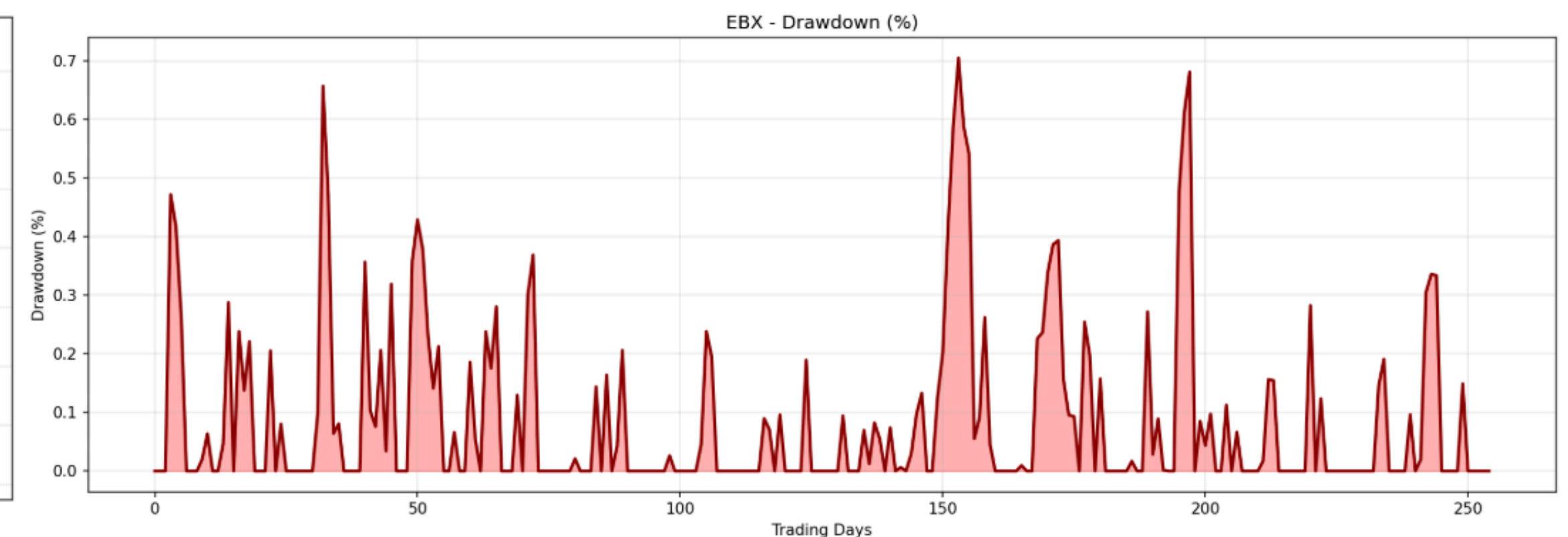
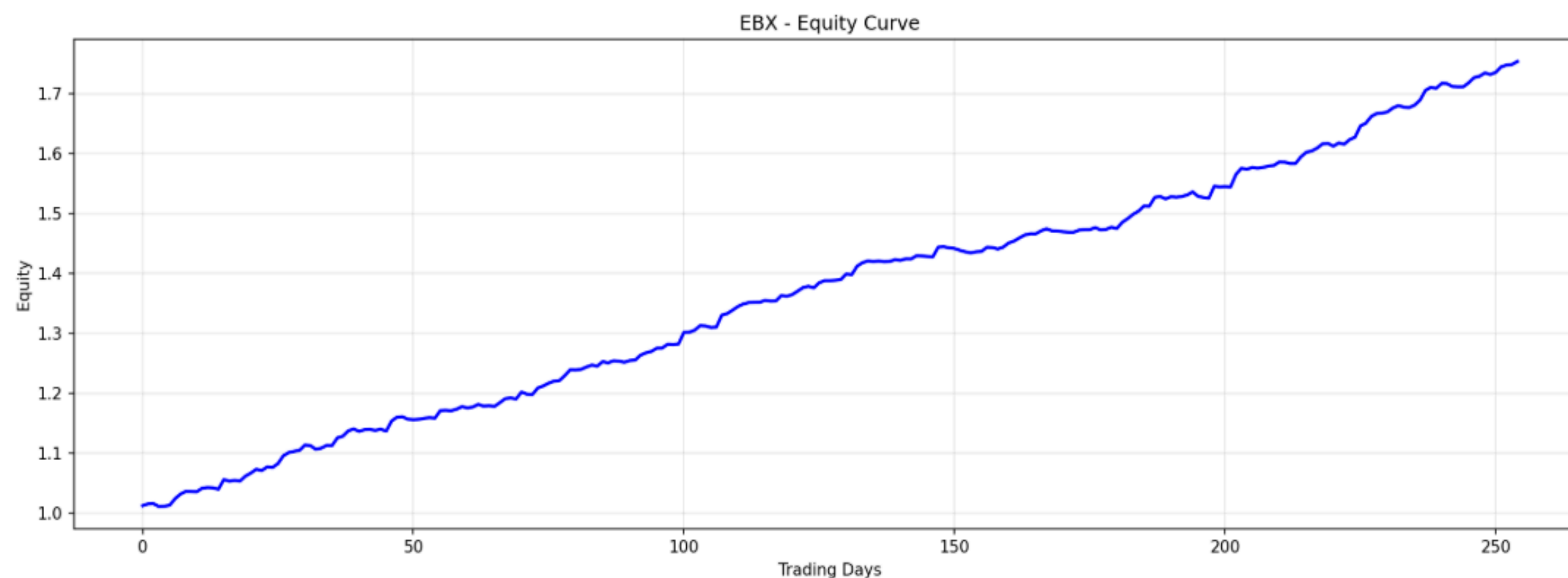
**7.30** Sharpe Ratio

**38.15** Sortino Ratio

**69.33%** Win Rate

**13.77** Avg. Trades/Day

## Equity And Drawdown Plots for 255 testing days



# Results and Key Metrics for EBY

**77.97%** Annualized Returns

**1.22%** Max Drawdown

**63.91** Calmar Ratio

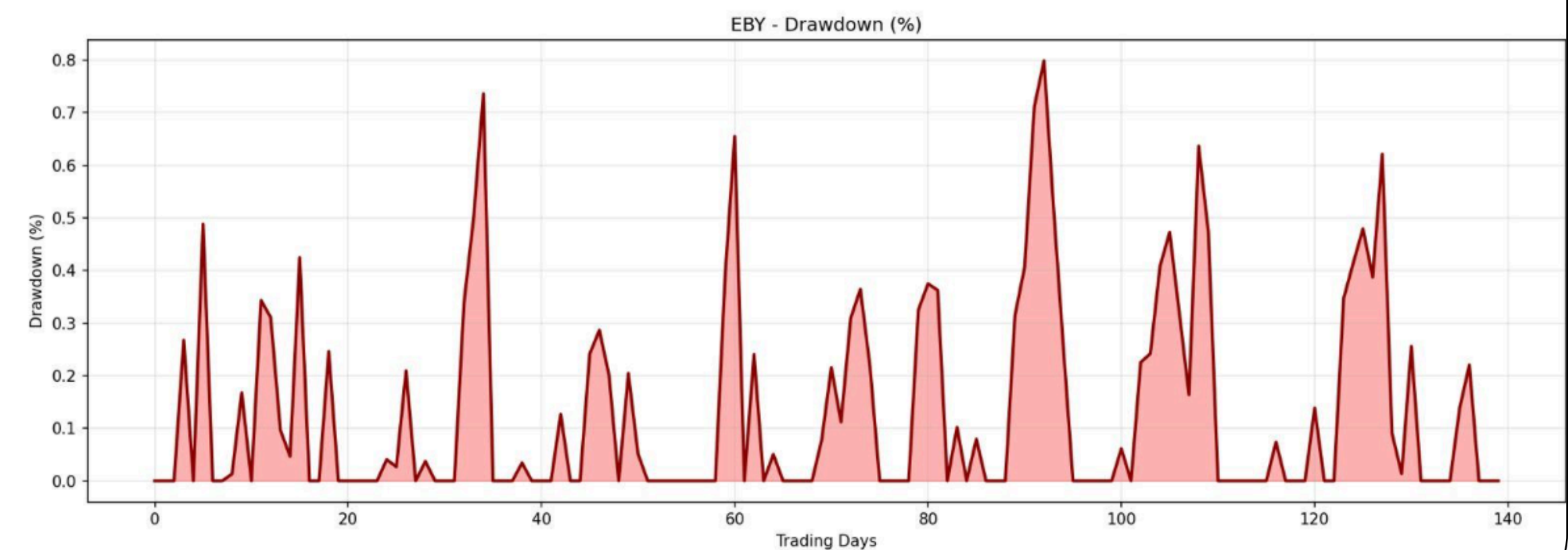
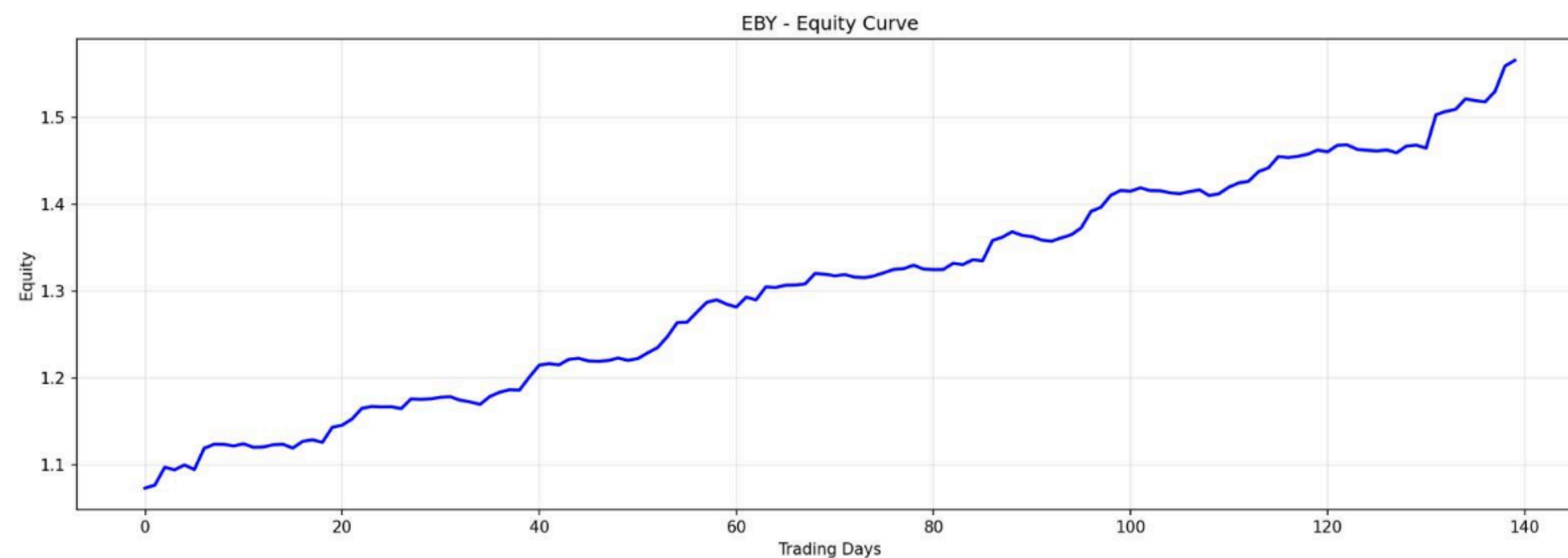
**4.91** Sharpe Ratio

**23.17** Sortino Ratio

**69.08%** Win Rate

**18.85** Avg. Trades/Day

## Equity And Drawdown Plots for 140 testing days





# EBX (255 Days)

01 PnL - 75.22 %  
Drawdown - 0.67 %  
Calmar - 110.26

06 PnL - 74.99 %  
Drawdown - 0.62 %  
Calmar - 118.89

02 PnL - 97.61 %  
Drawdown - 0.54 %  
Calmar - 177.48

07 PnL - 81.52 %  
Drawdown - 0.92 %  
Calmar - 88.02

03 PnL - 65.86 %  
Drawdown - 1.15 %  
Calmar - 56.38

08 PnL - 69.66 %  
Drawdown - 1.08 %  
Calmar - 63.62

04 PnL - 79.39 %  
Drawdown - 1.09 %  
Calmar - 72.01

09 PnL - 114.69 %  
Drawdown - 0.97 %  
Calmar - 116.57

05 PnL - 88.42 %  
Drawdown - 0.97 %  
Calmar - 90.07

10 PnL - 80.1- %  
Drawdown - 1.04 %  
Calmar - 75.91

# EBY (140 Days)

01 PnL - 43.00 %  
Drawdown - 1.22 %  
Calmar - 63.64

06 PnL - 41.96 %  
Drawdown - 1.00 %  
Calmar - 75.62

02 PnL - 41.18 %  
Drawdown - 1.19 %  
Calmar - 62.06

07 PnL - 45.77 %  
Drawdown - 1.10 %  
Calmar - 74.84

03 PnL - 37.69 %  
Drawdown - 1.07 %  
Calmar - 63.48

08 PnL - 37.51 %  
Drawdown - 1.06 %  
Calmar - 63.42

04 PnL - 39.76 %  
Drawdown - 1.07 %  
Calmar - 66.76

09 PnL - 44.23 %  
Drawdown - 1.13 %  
Calmar - 70.61

05 PnL - 45.77 %  
Drawdown - 0.89 %  
Calmar - 92.10

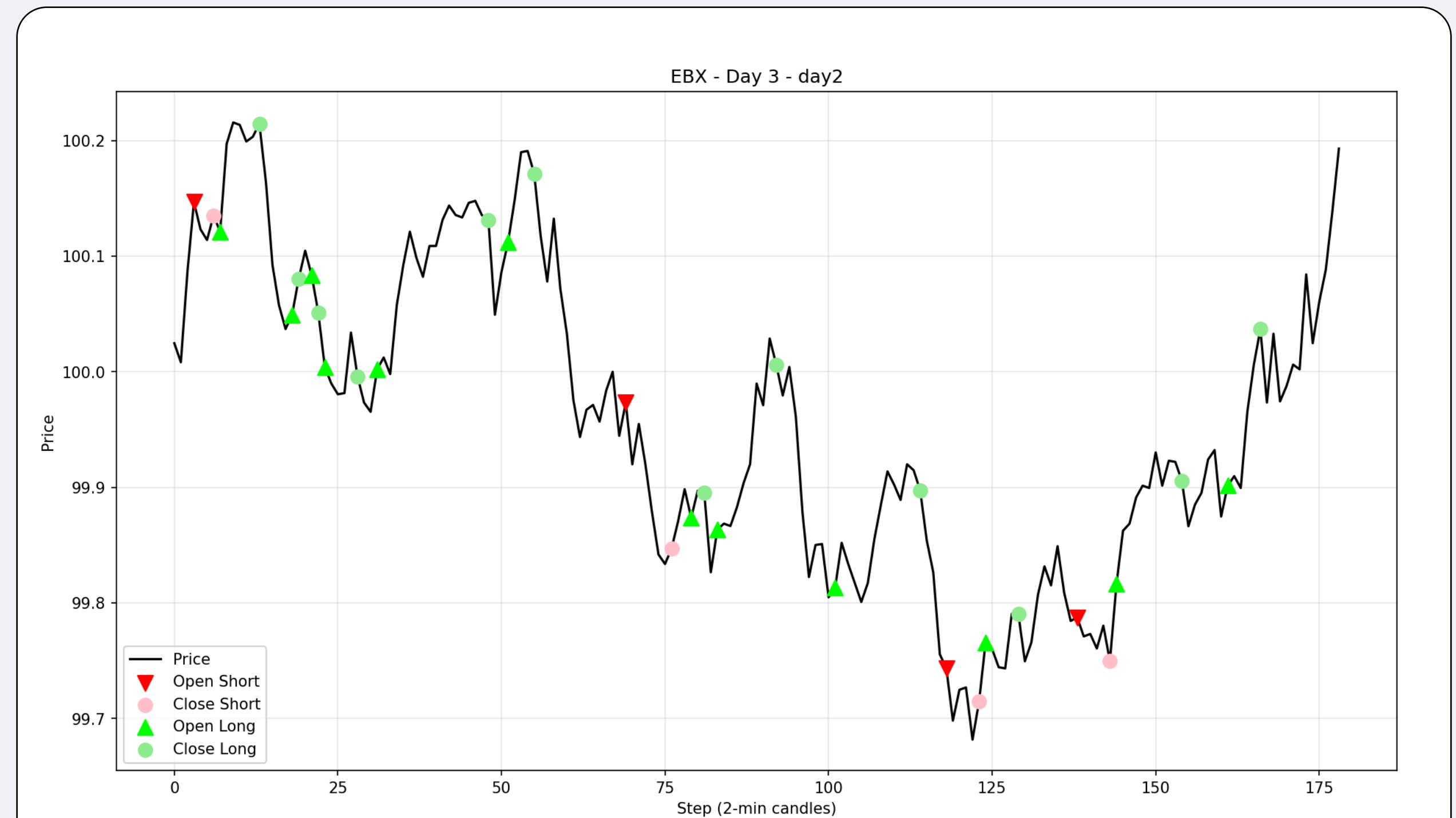
10 PnL - 56.31 %  
Drawdown - 0.83 %  
Calmar - 121.70

# Controlled Intraday Scalping

Frequent short-duration trades (~15/day) with hard downside control via **Stop Loss** and dynamic protection via **Trailing Stop Loss**.

## Risk Management

- Intraday markets produce repeated micro-moves.
- Our agent is designed to capture small directional edges quickly rather than hold long positions.
- Average price change between closing price of consecutive candles is ~**0.02%-0.03%**
- **Stop Loss: 0.04%** per position
- **Trailing Stop Loss: 0.04%** per position
- **End-of-day flattening:** No overnight exposure



- **Scalping clusters:** Multiple quick entries/exits
- **Risk exits:** SL = 0.04% prevents large drawdowns
- **TSL exits:** Profits protected when momentum fades

# Thank you.

“Don’t predict the future. Quantify it.”